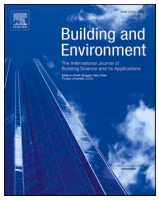
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Seeking comfort in the urban heat: Unraveling the effects of thermal-acoustic environments on psychological restoration in urban greenways

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ABSTRACT

As environmental pressures from high-density urbanization increase, the comfort of outdoor spaces and their restorative benefits for urban residents have become increasingly important. Although thermal and acoustic environments are key factors influencing outdoor comfort, evidence linking thermal and acoustic comfort with restorative benefits is scarce. This study explores the impact of the thermal-acoustic environment on psychological restorative benefits in urban greenways through on-site meteorological measurements and questionnaire surveys. The results indicate that, during the hot season, compared to densely vegetated greenways, sparsely vegetated greenways are perceived as hotter and noisier. Furthermore, the relationship between the thermal-acoustic environment and restorative benefits is more strongly correlated in sparsely vegetated greenways than in densely vegetated ones. In other words, the environmental comfort in different types of greenways can be targeted for improvement. Secondly, the interaction between thermal and acoustic environments influences restorative benefits. At the same sound pressure levels, restorative benefits decrease as the Universal Thermal Climate Index (UTCI) increases. However, on sparsely vegetated greenways, under conditions of 70–75 dB(A), the increase in UTCI does not significantly affect restoration. Finally, to identify the factors hindering psychological restoration, the study compares the veto power of thermal-acoustic comfort voting (TCV, ACV). This veto power varies according to greenway type. In densely vegetated greenways, ACV has greater veto power than TCV, while in sparsely vegetated greenways, the reverse is true. The findings offer valuable insights for urban planners and policymakers, suggesting that creating comfortable thermal-acoustic environments in outdoor spaces can enhance restorative benefits and contribute to the sustainability of high-density urban areas.

1. Introduction

The rapid pace of urbanization has intensified environmental pressures on residents living in high-density cities, such as noise pollution and heatwaves [1,2], contributing to increased rates of anxiety and cardiovascular diseases [3]. Consequently, the renewal and redevelopment of urban outdoor spaces have been increasingly employed to address the challenges posed by highly fragmented spatial patterns, gradually becoming a central issue in urban planning literature and practice worldwide [4]. The comfort of these spaces not only affects the psychological restoration of outdoor users but also contributes to the livability and sustainability of cities [5,6]. Among various environmental factors, thermal and acoustic conditions are primary determinants of the comfort of urban open spaces [7].

Urban green spaces, as an indispensable component of outdoor spaces, are becoming increasingly important and urgent due to their restorative benefits. On one hand, exposure to nature in cities can alleviate stress [8], while on the other, it supports outdoor recreational activities that promote positive emotions [9]. Urban greenways, a key type of urban green space [10], are particularly significant in high-density cities due to their multifunctionality [11]. They provide multidimensional benefits to cities, such as increasing the density of green areas within strict spatial constraints, offering public spaces for leisure and sports activities, and improving community quality of life [12]. However, in high-density cities in subtropical regions, the potential role of urban greenways in promoting environmental restoration

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may be underestimated. More importantly, as vital outdoor spaces for recreational activities, the thermal and acoustic environments of urban greenways may impact the psychological restoration of users [13]. As a result, urban planners and policymakers around the world are increasingly focused on maximizing the comfort of outdoor spaces with limited resources, whether through greenway planning or large-scale redevelopment [5]. To promote effective greenway design and renewal, planners must identify the key environmental comfort attributes that influence environmental restoration.

This study employs linear regression models and variance analysis applied to field data collected from urban greenways in Shenzhen, a high-density city in China, aiming to address three research questions:

- Research Question 1 (RQ1): What are the differences in thermal-acoustic perception between two greenway spaces with contrasting vegetation density?
- Research Question 2 (RQ2): How do thermal-acoustic environments influence psychological restoration across these two greenway spaces?
- Research Question 3 (RQ3): Do thermal and acoustic environments exert equal influence on psychological restoration in these greenway spaces?

This research makes threefold contributions to the literature on urban greenway planning and the practice of creating comfortable outdoor environments. First, it utilizes a unique field-based approach to investigate residents' perceptions of thermal and acoustic comfort in urban greenways within high-density environments. By integrating direct measurements of environmental conditions, the study provides nuanced, data-driven insights to guide practical interventions and inform sustainable outdoor design strategies. Second, the study explores the relationship between thermal-acoustic comfort and psychological restoration from a sensory interaction perspective. This perspective highlights how specific sensory conditions can disproportionately influence users' overall experiences, thus broadening planners' understanding of the complex interplay between environmental factors and psychological outcomes. The findings encourage further examination of multisensory dynamics in both theoretical frameworks and practical applications. Third, by analyzing variations in vegetation density within greenways, the study clarifies how different thermal-acoustic environments affect restorative outcomes. Addressing an under-examined aspect of urban greenway research, this work proposes an alternative to traditional approaches that rely solely on physical modifications. By explicitly connecting sensory experiences to psychological restoration, the results offer practical guidance for planners and policymakers seeking to enhance urban sustainability and improve quality-of-life outcomes through greenway design.

2. Literature review

2.1. The importance of urban greenways as outdoor spaces

Compared to traditional parks, urban greenways are often more suitable for recreational activities [14]. Their environmental design has become an important entry point for actively intervening in public health [15]. In recent years, the development of greenways has been vigorously promoted. As a form of small-scale linear green space, greenways play a crucial role for residents in high-density urban areas, providing essential spaces for daily activities [11]. They offer urban residents more convenient outdoor spaces and contribute to the creation of comfortable, livable environments [10]. Furthermore, high-quality urban greenways enhance exposure to the natural environment, thereby reducing perceived environmental stress and alleviating anxiety and depression [16], serving as restorative environments that promote both physical and mental well-being.

Research on restorative environments is largely anchored in two theoretical frameworks: Attention Restoration Theory (ART) and Stress Reduction Theory (SRT) [17,18]. According to SRT, exposure to natural settings helps alleviate stress and calm arousal, thereby promoting rapid recovery from acute stress [18]. In contrast, ART posits that non-directed attention activities—such as immersing oneself in nature—can restore directed attention, which is vital for tasks requiring focused effort [17]. While the restorative benefits of urban green spaces have been extensively documented, the specific attributes of greenways may influence their use through environmental perceptions [19], ultimately enhancing residents' well-being. Although the restorative effects of traditional, large-scale green spaces have been emphasized, ongoing urban land transformation and fragmented spatial patterns, coupled with the growing demand for accessible daily spaces [20], highlight the necessity and urgency of studying outdoor spaces—particularly greenways in high-density urban areas [11].

2.2. The impact of environmental comfort on psychological restoration

The comfort of outdoor environments has a positive influence on psychological well-being and restoration, a relationship that also informs urban design efforts [5,21]. However, much of the existing research on urban outdoor comfort has focused primarily on single-sensory perceptions [22,23], which may be insufficient for addressing environmental challenges in rapidly urbanizing contexts [24]. This narrow focus can also underestimate the role of sensory interaction [25,26]; by contrast, a multisensory experience is considered an effective intervention to enhance restorative outcomes [27]. In other words, multisensory perception is not merely the sum of individual sensory inputs but rather a composite effect that operates holistically [28]. For example, in summer, thermal comfort and acoustic comfort can interact, where high temperatures exacerbate acoustic discomfort, collectively influencing overall comfort [29]. Moreover, outdoor thermal comfort is often evaluated using the Universal Thermal Climate Index (UTCI) [30], owing to its robust accuracy and applicability in complex urban microclimates [31]. Meanwhile, the comfort of outdoor spaces is frequently assessed through their acoustic environment [32]. Specifically, the A-weighted equivalent sound pressure level (LAeq) is employed to characterize the intensity of ambient sound, and measures of acoustic comfort and preference are used to evaluate psychological impacts [33]. Consequently, there is a clear need to conduct a joint assessment of environmental comfort from both thermal and acoustic perspectives.

Although environmental comfort under multisensory interaction has been investigated [34], its relationship with psychological restoration in outdoor spaces remains largely unexplored. In addition, while acoustic comfort may enhance restorative benefits [35], variations in acoustic type and loudness under the same temperature conditions can shift thermal comfort evaluations [36,37]. These insights underscore the interactive effects of thermal and acoustic environments, but often overlook whether each factor contributes equally to perceived restoration. Prior research has introduced the notion of veto power—a situation in which a single severely negative stimulus, such as excessive noise or extreme heat, can override otherwise favorable conditions and substantially reduce overall satisfaction [25], prompting the question of whether a similar veto effect exists in the thermal-acoustic environment with respect to psychological restoration. Moreover, current research on thermal-acoustic comfort predominantly takes place in laboratory settings [37], potentially restricting our understanding of how complex environmental factors and psychological perceptions intertwine in green spaces. This gap is particularly pressing in high-density urban contexts, where visitors to green spaces encounter more severe environmental conditions, which may obscure potential links between environmental comfort and psychological restoration.

3. Methods

3.1. Site selection

This study conducted a field experiment in Shenzhen, China. Shenzhen's greenways have been developed alongside existing urban roads, forming a mature network that connects public green spaces and service facilities. Characterized by extensive coverage and flexible configurations [15], this network makes Shenzhen an ideal hotspot for research. Moreover, Shenzhen is not only representative of high-density urban environments but also experiences hot, humid summers, with average daily temperatures typically exceeding 30 °C [38]. As such, the city's greenways offer a compelling context in which to investigate the relationship between outdoor thermal-acoustic comfort and psychological restoration.

In selecting specific greenways, we relied on the Shenzhen Greenway Atlas published by the Shenzhen Municipal Administration and Comprehensive Enforcement Bureau, along with local greenway web maps and on-site surveys. Ultimately, greenways in the Nanshan District were chosen as study sites. While both selected sites fall under the official category of community greenways as defined in the Shenzhen Greenway Atlas, we further classified them from an urban greening perspective into two categories based on vegetation density: densely vegetated and sparsely vegetated greenways (Fig. 1). This vegetation-based distinction complements earlier functional classifications (e.g., landscape-recreational vs. service-oriented) proposed in previous studies [15], and captures differences in ecological exposure and spatial enclosure relevant to users' thermal-acoustic experiences. Densely vegetated greenways are typically located along sidewalks adjacent to urban parks, situated near major roads, and distributed in relatively discrete segments. They primarily serve as important recreational and exercise spaces for local residents. In contrast, sparsely vegetated greenways usually follow sidewalks along secondary or local streets, and are embedded within mixed-use urban settings that combine shopping, commuting, and leisure functions. Despite differences in immediate surroundings and vegetation density, both types belong to the broader urban greenway system designed to support non-motorized mobility and public open space access. Their inclusion enables a focused comparison of how variations in vegetation influence user experiences and perceptual outcomes within greenway spaces. Both greenways featured similar pedestrian surfaces, including asphalt, poured concrete, and concrete pavers commonly used in urban sidewalk construction.

3.2. Procedures

In this outdoor experiment, conducted between late June and early July (summer), participants were first briefed on the study's purpose and procedures, and asked to sign a consent form for data collection. At



Fig. 1. Location map of the research site.

the selected sites, participants reported their evaluations of comfort (Section 3.3.1) and psychological restoration (Section 3.3.2) while on the greenways. Simultaneously, thermal and acoustic parameters were measured (Section 3.3.3). Data were collected from 8:00–11:00 a.m. to 2:00–5:00 pm. to capture both morning and afternoon conditions while avoiding midday extremes, thereby minimizing the potential masking effect of heat stress on other sensory perceptions.

For thermal assessment, the UTCI was adopted to evaluate the physiological effects of the thermal environment [6,29]. Owing to UTCI's wide acceptance and robust methodology in previous studies, it was used to represent thermal comfort [38], with corresponding stress categories based on specific UTCI ranges [39]. The thermal stress categories used in this study—very strong heat stress (+38 to +46 °C), strong heat stress (+32 to +38 °C), and moderate heat stress (+26 to +32 °C)—follow internationally recognized UTCI thresholds. These classifications have been widely applied in both cold and subtropical climates [6,38]. As for the acoustic environment, the A-weighted equivalent continuous sound level (LAeq) was selected to account for the intermittent nature of traffic noise, thereby serving as a suitable metric for assessing the greenway sound environment. Following the categories used in prior studies [29], LAeq was classified into four ranges: 55–60 dB(A), 60–65 dB(A), 65–70 dB(A), and 70–75 dB(A). Field observations confirmed that the main sound sources at this site were urban traffic, human activity, and occasional mechanical noise, with intermittent natural sounds such as wind in the vegetation and birdsong. No sea-related sounds were perceptible.

Following the experiment, we randomly selected 47 participants—each of whom had completed the questionnaire and provided verbal consent—for brief, on-site interviews. Conducted outdoors and lasting 3 to 5 min, the semi-structured interviews explored participants' perceptions of thermal comfort and acoustic disturbance, with questions including, for example, "what would make this greenway feel more comfortable to you?" and "were there any sounds that made you feel uncomfortable?" (Fig. 2).

3.3. Measures

3.3.1. Assessment of thermal and acoustic comfort

In this study, we assessed thermal-acoustic comfort in the selected greenway environments by designing a questionnaire informed by ISO 10551 guidelines, ASHRAE Standards [40], and previous research on evaluating outdoor thermal and acoustic environments [25,29]. Specifically, the questionnaire included the Thermal Sensation Vote (TSV), the Thermal Comfort Vote (TCV), the Acoustic Sensation Vote (ASV), and the Acoustic Comfort Vote (ACV) (Appendix A). TSV captures consciously perceived thermal sensation, while ASV measures perceived loudness. TCV and ACV reflect satisfaction with the thermal and acoustic conditions, respectively. TCV and ACV were assessed using a 7-point bipolar Likert scale with a neutral midpoint, representing a state of neither satisfaction nor dissatisfaction, a format widely adopted in outdoor environmental comfort research to enable nuanced measurement of comfort perceptions in field settings [25] (Table 1).

Table 1
Subjective evaluations of the questionnaire surveys.

Score	Thermal Sensation Vote	Thermal Comfort Vote	Acoustic Sensation Vote	Acoustic Comfort Vote
3	Hot	Very comfortable	Very quiet	Very comfortable
2	Warm	Comfortable	Quiet	Comfortable
1	Slightly warm	Slightly comfortable	Slightly quiet	Slightly comfortable
0	Neutral	Neutral	Neutral	Neutral
−1	Slightly cool	Slightly uncomfortable	Slightly loud	Slightly uncomfortable
−2	Cool	Uncomfortable	Loud	Uncomfortable
−3	Cold	Very uncomfortable	Very loud	Very uncomfortable

3.3.2. Measurement of psychological restorative benefits

The Restorative Outcome Scale (ROS) was used to measure participants' level of psychological restoration. This scale has been widely employed and validated in studies examining urban outdoor environments and psychological restoration [41,42]. It consists of six items [43]: one item assesses attentional recovery (e.g., "my concentration and alertness clearly increase here"), two items address stress relief (e.g., "I can forget everyday worries here"), and three items evaluate mental relaxation and calm (e.g., "after visiting this place, I always feel restored and relaxed"). Each item is rated on a five-point Likert scale, ranging from "disagree strongly" (−2) to "agree strongly" (+2). A composite score is obtained by averaging all six items, representing the overall restorative benefit.

3.3.3. Measurement of environmental parameters

During the field experiment, the thermal and acoustic conditions were recorded in parallel. The thermal parameters measured included air temperature (*Ta*), relative humidity (*RH*), wind speed (*Va*), and globe temperature (*Tg*). The acoustic parameter was the A-weighted equivalent continuous sound pressure level (*LAeq*). A portable Kestrel 5500 weather station was used to record wind speed and direction, while the BES-01 temperature recorder measured *Tg*. The black globe measured 0.08 m in diameter, and its surface material had a scattering coefficient of 0.95. A BES-02 temperature and humidity recorder measured air temperature and humidity. All devices were placed under a radiation-resistant aluminum hood to minimize interference from direct solar radiation and wind. To assess the acoustic environment, an NC-75 acoustic calibrator was employed to calibrate the NL-52A sound level meter, ensuring accurate measurements. All instruments were positioned 1.5 m above the ground. Table 2 provides further details on the recorded parameters.

3.4. Participants

This experiment was conducted on both weekdays and weekends from late June to early July. A total of 381 individuals were initially approached for participation. Among them, 27 declined due to time constraints, and 12 did not complete the full experimental protocol. As a

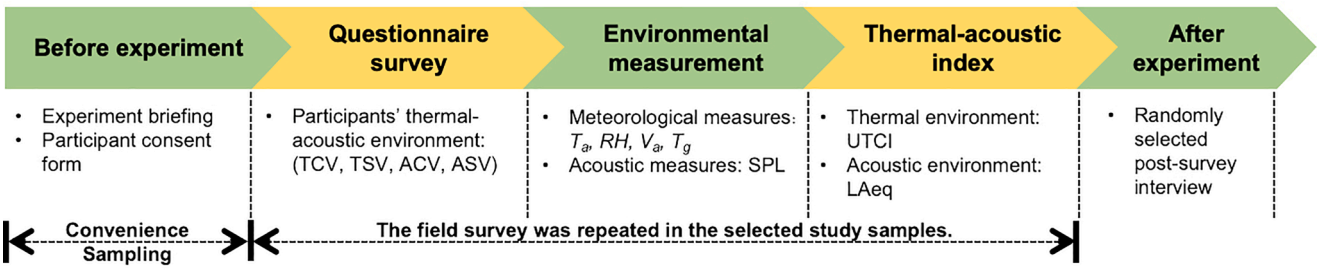


Fig. 2. Experimental procedure.

Table 2
Details of the measurement instruments.

Instrument	Parameters	Range	Precision
Kestrel 5500 weather station	Wind speed	0.4~40 m/s	±0.1 m/s
BES-01 temperature recorder	Air temperature	−30–50 °C	±0.5 °C
BES-02 temperature and humidity recorder	Relative humidity	0 −90 %RH	±3 %
NL-52A Sound Level Meter	Sound pressure level	25 dB(A)~130 dB(A)	±1 dB

result, 342 participants were successfully recruited through convenience sampling (56.14 % male; mean age = 35, SD = 8). A total of 153 participants were ultimately recruited at densely vegetated urban greenways (DVUG) and 189 participants at sparsely vegetated urban greenways (SVUG), based on the two greenway types classified in Section 3.1. Additionally, participants had primarily been walking, standing, or sitting during the 20 min preceding the experiment. The selected greenway segments ranged between 300 and 500 m in length. Participants were instructed to walk naturally along the designated greenway path and, after exposure, to rest briefly at a fixed point while completing the questionnaire. Although the walking distance contributed to the contextual experience of the site, all perceptual responses were recorded while participants were in a stationary state, ensuring consistency in exposure conditions and minimizing variation related to individual movement. As the study was conducted in summer, the vast majority of participants wore typical summer clothing (upper body: short-sleeve T-shirt; lower body: shorts, short skirts, or long skirts; accessories: hats or facemasks).

Prior to the start of the experiment, each participant was required to rest quietly for 5 min to allow perceptual stabilization. All participants confirmed they were not experiencing mental fatigue or negative emotions and retained full cognitive and decision-making capacity [44]. Participants were then exposed to the greenway environment for approximately 10 min, followed by the completion of the questionnaire, which took about 3 min. Previous research has indicated that such an exposure duration is sufficient to elicit perceptual and psychological restorative effects in outdoor settings. To ensure the accuracy of the experiment, all participants confirmed that they were not experiencing mental fatigue, had not consumed any stimulants (e.g., caffeine or nicotine). Before the experiment, participants were informed about the study procedures, and they were assured that all questionnaire data would be used exclusively for academic research purposes. Additionally, confidentiality was strictly maintained, and no personal information was disclosed under any circumstances.

3.5. Data analysis

First, analysis of variance (ANOVA) was employed to examine whether there were significant differences in thermal-acoustic perceptions across different types of greenways, as well as to identify the role of thermal-acoustic environments in restorative outcomes. Second, weighted linear regression models were used to explore the relationships between ROS scores and the measured UTCI and LAeq values. To facilitate analysis, all data were categorized by greenway type, and within each category, data points were grouped in increments of 1 °C UTCI, after which the mean values were computed. Subsequently, a two-way ANOVA was conducted to investigate the combined effects of thermal and acoustic conditions on psychological restoration. Finally, conditional probability analysis was utilized to test for potential "veto effects," wherein the interaction of thermal and acoustic comfort could significantly affect psychological restoration. All statistical analyses were performed using SPSS version 26.0.

4. Results

4.1. Comparison of thermal-acoustic comfort across different greenway types

To answer RQ1, we conducted analysis of variance to compare thermal-acoustic perceptions between different types of greenways: densely vegetated and sparsely vegetated. Participants reported significantly higher levels of comfort in densely vegetated greenways compared to sparsely vegetated ones, evident in both TCV (densely: 0.97 vs sparsely: 0.28; $p < 0.01$) and ACV (densely: 1.33 vs sparsely: 0.36; $p < 0.01$). Conversely, compared to densely vegetated greenways, participants perceived sparsely vegetated greenways as hotter (TSV, densely: 0.70 vs sparsely: 1.47; $p < 0.01$) and noisier (ASV, densely: 0.77 vs sparsely: −0.47; $p < 0.01$) (Fig. 3).

4.2. Impact of thermal environment on psychological restoration

Fig. 4 illustrates the relationship between UTCI and restorative outcomes. In both densely vegetated ($R^2 = 0.552$) and sparsely vegetated ($R^2 = 0.688$) greenways, UTCI exhibits a moderate negative correlation with ROS. This indicates that regardless of greenway type, an increase in UTCI is associated with a gradual decline in restorative benefits. Moreover, this trend is more pronounced in sparsely vegetated greenways.

To further investigate the impact of thermal comfort votes (TCV) on ROS under different thermal stress conditions, an analysis of variance was conducted. The results indicate that under moderate heat stress, strong heat stress, and very strong heat stress conditions, TCV significantly influenced ROS in both densely vegetated and sparsely vegetated greenways. Notably, ROS evaluations were higher in densely vegetated greenways. Fig. 5 illustrates the relationship between TCV and ROS. When TCV ≥ -1, restorative benefits were only absent under very strong heat stress conditions. However, when TCV ≥ -2, despite thermal discomfort in the densely vegetated greenway, ROS still exhibited restorative benefits under moderate heat stress. Across all thermal stress conditions, ROS increased with rising TCV. In sparsely vegetated greenways, the differences in ROS between thermally comfortable and uncomfortable visitors were 2.42 under very strong heat stress, 2.34 under strong heat stress, and 2.61 under moderate heat stress. In contrast, in densely vegetated greenways, ROS differences remained relatively stable across thermal stress conditions, averaging around 2.3. This suggests that the influence of TCV on ROS is stronger in sparsely vegetated greenways. These findings provide an initial response to RQ2.

4.3. Impact of acoustic environment on psychological restoration

Fig. 6 illustrates a moderate negative correlation between LAeq and ROS. Specifically, the relationship between LAeq and ROS is more pronounced in densely vegetated greenways ($R^2 = 0.622$) compared to sparsely vegetated greenways ($R^2 = 0.536$). This suggests that the acoustic environment has a greater impact on ROS in densely vegetated greenways than in sparsely vegetated ones. As LAeq increases, restorative benefits gradually decline.

Fig. 7 illustrates the relationship between ACV and restorative benefits under different acoustic conditions. ANOVA comparisons between greenway types reveal a significant difference in the impact of ACV on restoration. In densely vegetated greenways, even when ACV ≥ -2, ROS continues to exhibit restorative benefits at 55–60 dB(A). However, this effect is not observed in sparsely vegetated greenways, where ROS only demonstrates restorative benefits when ACV ≥ -1 under 55–60 dB(A) conditions. Regarding the influence of ACV on ROS, at the same LAeq level, ROS significantly increases as ACV improves. The average range (i.e., the difference between the maximum and minimum values) was 1.71 in densely vegetated greenways and 1.86 in sparsely vegetated greenways. Concerning the impact of the acoustic environment on ROS, at the

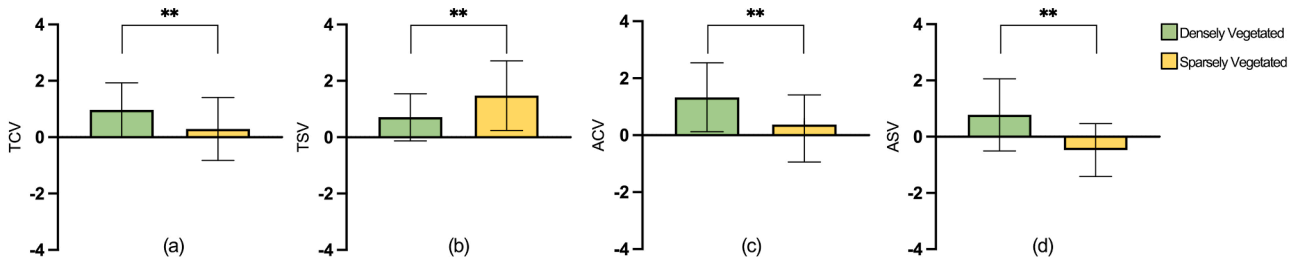


Fig. 3. Comparison of thermal and acoustic perceptions across different greenway types; (a) thermal comfort comparison; (b) thermal sensation comparison; (c) scoustic comfort comparison; (d)scoustic sensation comparison.

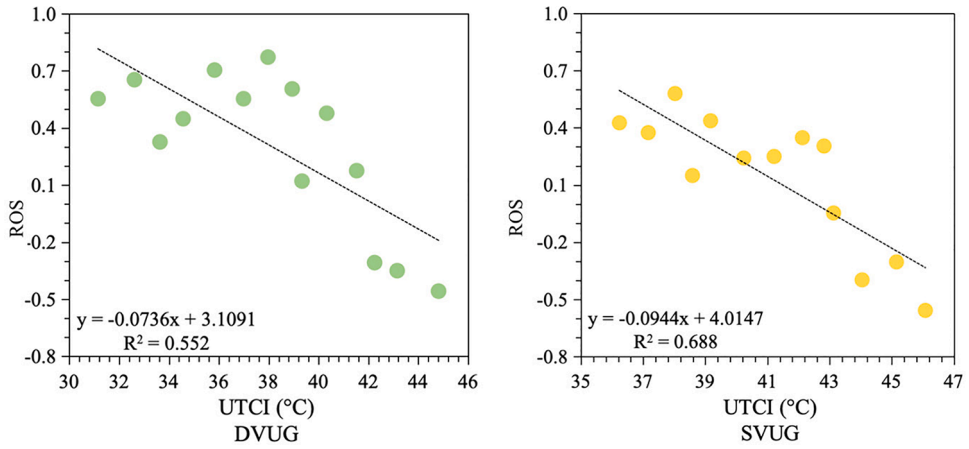


Fig. 4. Correlation between UTCI and ROS (based on 1 °C UTCI bins).

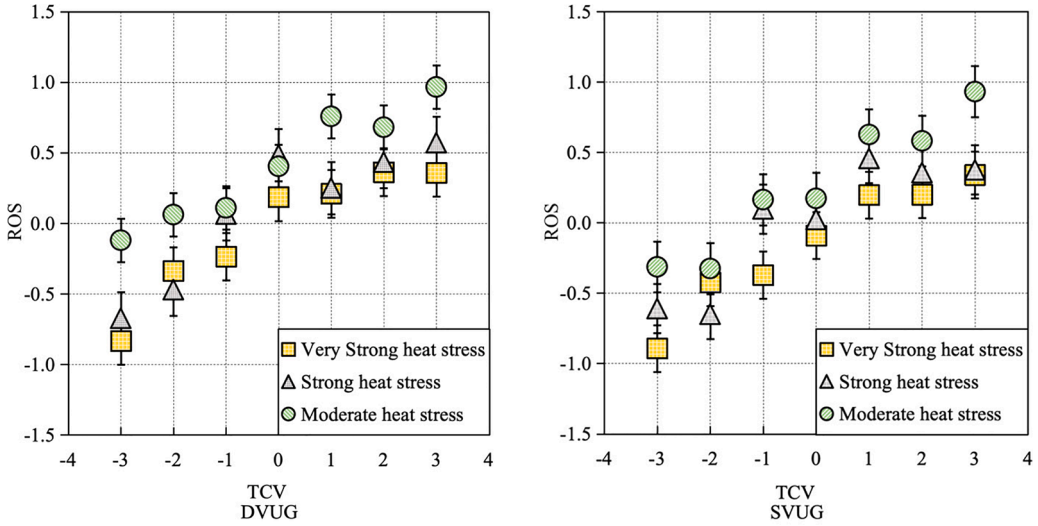


Fig. 5. Relationship between thermal comfort votes and psychological restoration.

same ACV level, ROS exhibits a decreasing trend with increasing LAeq in sparsely vegetated greenways. Specifically, when LAeq ranges from 55 to 60 dB(A) to 70–75 dB(A), the average ROS difference is 0.86, further suggesting that higher noise levels in sparsely vegetated greenways may reduce restorative benefits. This provides a more detailed response to RQ2.

4.4. Combined effects of thermal-acoustic interactions on psychological restoration

We examined the interaction effects of UTCI and LAeq on ROS across different types of urban greenways. As shown in Fig. 8, both environmental variables demonstrated interactive influences on ROS in both sparsely and densely vegetated greenways. First, regarding thermal effects, ROS generally decreased with increasing UTCI under the same LAeq levels in sparsely vegetated greenways. However, when LAeq reached 70–75 dB(A), the influence of UTCI on ROS became less pronounced. Second, in terms of acoustic influences, ROS consistently declined as LAeq increased under each UTCI level in both greenway types. Moreover, the average fluctuation of ROS—calculated by comparing the mean ROS across all LAeq categories between higher and lower UTCI levels—was 0.62 in densely vegetated greenways and 0.73

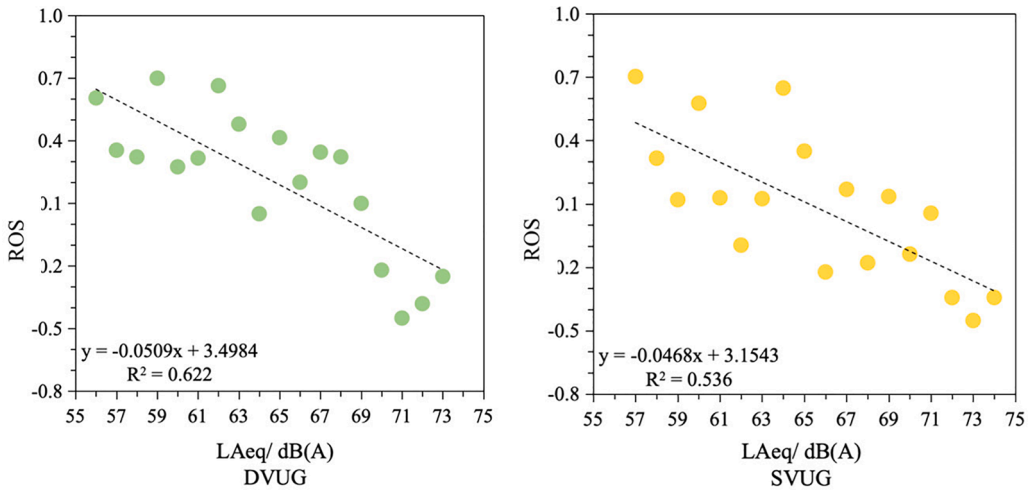


Fig. 6. Correlation between LAeq and ROS.

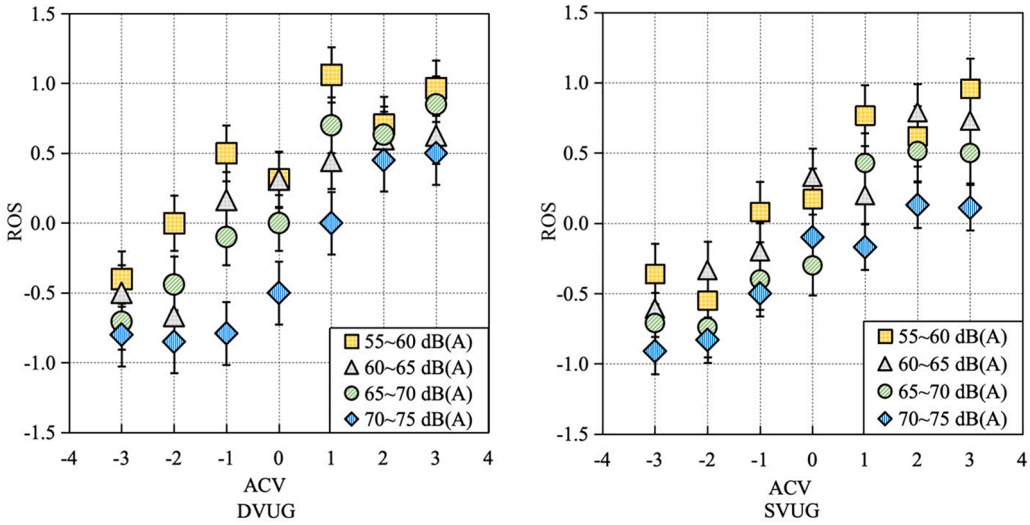


Fig. 7. Relationship between acoustic comfort votes and psychological restoration.

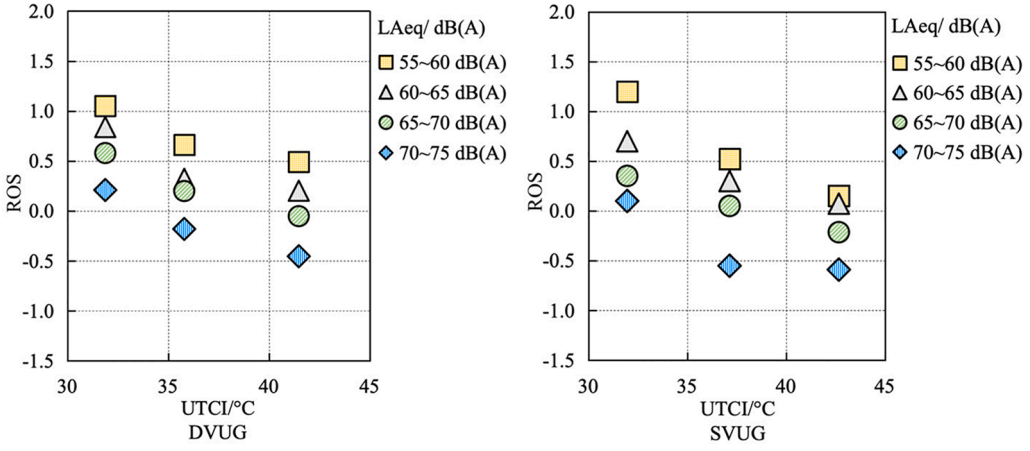


Fig. 8. Effect of thermal-acoustic interaction on ROS.

in sparsely vegetated ones. This suggests that densely vegetated greenways offer more consistent and stable restorative experiences in response to thermal variation, reflecting greater psychological resilience. Finally, the range of ROS—defined as the difference between the maximum and minimum ROS values across UTCI levels within each LAeq category—varied with noise exposure. In densely vegetated greenways, this range remained relatively small (from 0.55 to 0.66), suggesting that thermal stress had a limited impact on restorative

perception in this setting. This highlights the relatively greater role of acoustic conditions in supporting perceived restoration under high vegetation cover. In contrast, in sparsely vegetated greenways, the ROS range decreased notably from 1.04 to 0.69 as LAeq increased, indicating that thermal conditions played a more substantial role, particularly under quieter conditions. This suggests that improving thermal comfort may be more effective in enhancing restoration in sparsely vegetated greenways. This provides a comprehensive answer to RQ2.

4.5. Influence of thermal-acoustic comfort on psychological restoration

To address RQ3, we employed conditional probability analysis to elucidate the relationship between thermal-acoustic comfort and psychological restoration [25]. Conditional probability refers to the likelihood of one event occurring in the context of another event that is already known or assumed to have taken place. It is typically denoted as P(A | B), representing the probability of event A under the condition that event B has occurred. Specifically, we calculated probabilities to describe the likelihood of experiencing "adverse psychological restoration" (APR) under different combinations of comfort factors in greenway environments with varying vegetation densities. First, we defined probability events representing outdoor comfort votes (Table 3). Based on previous research, comfort votes within the range of [−3,0] were classified as "uncomfortable," while psychological restoration scores within [−2,0] were defined as "APR" [5]. This classification resulted in a total of $2^2 = 4$ possible comfort condition combinations, as outlined in Table 3.

Table 4 presents the probability of different comfort vote combinations occurring under APR conditions, allowing for an examination of the veto effect of comfort perceptions on psychological restoration. The combinations were defined as follows: $S_1 = (X_1,Y_1)$, $S_2 = (X_1, Y_2)$, $S_3 = (X_2, Y_1)$, $S_4 = (X_2, Y_2)$. Fig. 9 illustrates the proportion of APR occurrences within various thermal-acoustic discomfort vote combinations. For instance, in the sparsely vegetated urban greenway (SVUG) category, $n = 2$ represents greenway users who felt "comfortable" in terms of thermal comfort but "uncomfortable" in terms of acoustic comfort, with an APR probability of 25.18 %. Additionally, even when users reported comfort in both thermal and acoustic conditions, they could still experience APR. This occurred more frequently in SVUG compared to DVUG, with an incidence rate of 15.12 %. Notably, acoustic discomfort exhibited a stronger veto effect on psychological restoration in densely vegetated greenways (32.50 %), whereas thermal discomfort showed a higher veto tendency in sparsely vegetated greenways (30.35 %). When both thermal and acoustic comfort votes were rated as "uncomfortable," the likelihood of experiencing APR was significantly higher in SVUG (73.62 %).

5. Discussion

5.1. Interaction between thermal and acoustic environments

The interaction between thermal and acoustic environments significantly influences the restorative benefits of different types of greenways. Regarding the impact of the thermal environment on ROS in densely vegetated greenways, the effect of LAeq on ROS varies across different UTCI levels. As UTCI increases from 31.7 °C to 41.4 °C, ROS evaluations decline with rising LAeq, demonstrating a downward trend within the same UTCI category. This suggests that when summer temperatures are excessively high and thermal stress is strong, ROS is significantly affected, decreasing as LAeq increases. However, this trend is relatively moderate within the 55–60 dB(A) range but becomes more pronounced under 70–75 dB(A) conditions. In other words, even under high thermal stress, a favorable acoustic environment in densely vegetated greenways provides an opportunity to mitigate its negative effects on restoration. This finding aligns with previous research suggesting that acoustic comfort can enhance individuals' tolerance to thermal conditions, thereby fostering more positive perceptual experiences [45]. For instance, vegetation with different leaf morphologies can help reduce urban noise [46], offering a sustainable and easily implementable alternative to constructing noise barriers while also contributing to urban greenery. Such approaches provide practical design solutions for addressing thermal exposure in outdoor environments. These findings support the notion that poor performance in a single sensory dimension may obscure the influence of multisensory interactions. Therefore, it is crucial to consider multisensory interactions in order to help policymakers and urban planners identify appropriate intervention strategies to optimize the restorative potential of various outdoor spaces.

Furthermore, another key contribution of this study is its deeper insight into the factors contributing to "unfavorable" psychological recovery. In hot climates, thermal exposure can negatively impact mental

Table 3
Descriptions of probability conditions.

Probability events	Notations	Explanations
Psychological restoration	A	Adverse to psychological restoration
Acoustic comfort vote	Y1	Comfortable with acoustic environment
	Y2	Uncomfortable with acoustic environment
Thermal comfort vote	X1	Comfortable with thermal environment
	X2	Uncomfortable with thermal environment

Table 4
Outcomes of conditional probability analysis.

		P(A/Sn)	
		DVUG	SVUG
$n = 1$	S1=(X1,Y1)	0.0842	0.1512
$n = 2$	S2=(X1,Y2)	0.3250	0.2518
$n = 3$	S3=(X2,Y1)	0.1638	0.3035
$n = 4$	S4=(X2,Y2)	0.6825	0.7362

Note: P(A | Sn) denotes the probability of APR occurring under thermal-acoustic comfort condition Sn. Each Sn represents a unique combination of thermal (X) and acoustic (Y) comfort states. DVUG refers to densely vegetated urban greenway; SVUG refers to sparsely vegetated urban greenway.

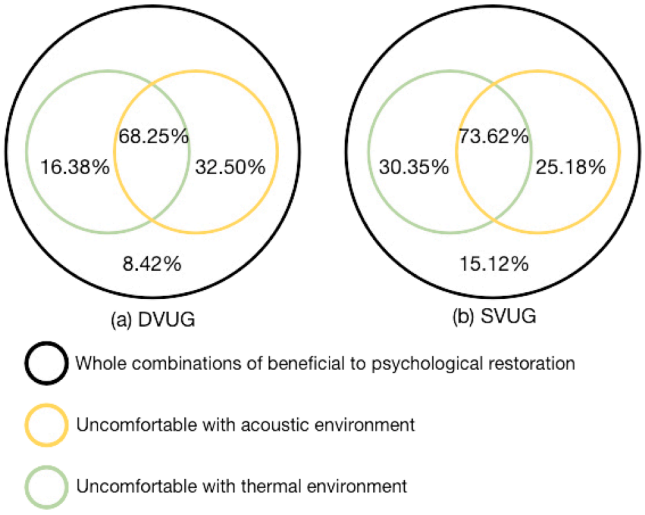


Fig. 9. Distribution of adverse psychological restoration by thermal-acoustic comfort conditions. (Black circles represent all combinations that contribute to psychological restoration. The numbers indicate the probability of APR occurrence based on comfort vote combinations. Within the green and yellow circles, areas denote "uncomfortable" TCV/ACV conditions, while their overlap signifies cases where both comfort votes are "uncomfortable." In contrast, the black circle, excluding the green and yellow areas, represents conditions where both thermal and acoustic comfort votes are "comfortable.").

well-being, an effect that is particularly pronounced in high-density urban areas [47]. In highly vegetated outdoor environments, ACV has a stronger impact on psychological restoration than other sensory comfort factors [48], as individuals may be more sensitive to noise in thermally stressful conditions [7]. This suggests that in less vegetated environments, people may tolerate situations where they feel acoustically uncomfortable but thermally comfortable. For instance, in sparsely vegetated settings, visitors are more sensitive to thermal comfort, with thermal comfort demonstrating a stronger veto effect (30.35 %) on ROS in these greenways.

5.2. Impact of environmental conditions on restoration

First, this study extends previous research on the influence of landscape and soundscape on subjective perceptions of restorative benefits in outdoor spaces by further exploring the role of thermal and acoustic environmental conditions. It further reveals the interrelationship between thermal and acoustic environments based on their physical conditions. As UTCI and LAeq increase, ROS evaluations gradually decline. Regarding the impact of the thermal environment on ROS, the effect of TCV varies under different levels of thermal stress. When individuals are dissatisfied with the thermal environment, ROS is lower. This finding supports the correlation between UTCI and ROS rather than just temperature alone [6]. This relationship can be explained by the influence of solar radiation and subjective appraisal of thermal comfort [49], which reflects both physical exposure and psychological judgment.

Second, concerning the impact of the acoustic environment on ROS, we found that under different acoustic conditions, respondents who perceived a comfortable sound environment reported higher ROS levels. This effect was particularly evident in greenway settings. In both densely vegetated and sparsely vegetated greenways, even when ACV was rated as slightly uncomfortable (ACV ≥ -1), psychological restoration was still observed under sound environments ranging from 55 to 60 dB(A) and 60 to 65 dB(A). This phenomenon can be explained by the restorative potential of preferred soundscapes [50], where natural sounds are often more accepted [51], especially in urban greenways. Additionally, ACV may be influenced by individual subjective preferences and perceptual sensitivity [52], which should not be overlooked in assessing its impact on psychological restoration.

Finally, this study expands previous research on restorative environments based on Attention Restoration Theory and Stress Recovery Theory, emphasizing that environmental restorative benefits are significantly influenced by thermal and acoustic conditions, rather than being solely dependent on visual factors [18]. By incorporating objective measurements of the thermal and acoustic environment, this study complements subjective assessments of comfort and helps explain how perceived discomfort under different environmental conditions may translate into changes in restoration outcomes [21]. This approach provides a more scientific and practical foundation for the design and application of outdoor spaces.

5.3. Implications for urban greenway design

To support more targeted greenway planning, this study integrates both subjective comfort evaluations and objective environmental indicators. While subjective ratings capture individual experiences of comfort, objective metrics help identify the physical conditions shaping those perceptions. They not only clarify the relative influence of thermal and acoustic factors, but also reveal where landscape interventions—such as shading or vegetation buffers—may be most effective. This integrated approach acknowledges that comfort is shaped by both environmental conditions and psychological adaptation, and it enables more responsive and evidence-based design decisions.

Although research has demonstrated the numerous benefits of natural environments for physical and mental well-being, urban greenways—as typical outdoor spaces—face dual challenges. First, they are often underutilized in hot weather, and second, the combined effects of soundscape and climate on restorative benefits in high-density urban environments remain insufficiently explored. In particular, during hot seasons and under noisy conditions, subjective discomfort may discourage people from using outdoor spaces. These findings highlight the importance of addressing thermal and acoustic discomfort in outdoor environments to maximize their restorative potential. Urban street vegetation can enhance outdoor thermal comfort, mitigate noise, and alleviate stress [18,46,53]. For example, clusters of trees provide better noise reduction than individual trees, and dense greenery offers shade that reduces solar radiation, lowers local temperatures, and enhances overall comfort. To improve outdoor spatial comfort, landscape designers and policymakers should prioritize these strategies.

Empirically, our findings offer guidance for differentiated greenway design. In densely vegetated streets, regulating the sound environment—such as using vegetation to reduce noise and create a more natural, comfortable soundscape—emerges as a key strategy for enhancing restorative benefits, as soundscape influences are more pronounced in such settings. Conversely, in sparsely vegetated greenways, thermal conditions play a greater role in restoration. Therefore, implementing artificial shading structures (e.g., canopies and street-scale shelters) to reduce temperatures is an effective intervention [54]. As urban greenways serve as vital outdoor spaces in high-density cities, their spatial configuration and design should emphasize comfort. In areas with high usage during hot seasons, shade should be prioritized to facilitate outdoor activities in thermally comfortable microclimates. These measures not only promote physical and mental restoration but also reduce dependence on energy-intensive cooling systems [38], contributing to sustainable development goals.

5.4. Limitations and prospects

This study conducted field-based assessments of the thermal and acoustic environments of urban greenways during the hot summer season in a high-density city. By quantifying visitors' subjective evaluations of environmental comfort and psychological restoration, the research contributes to a deeper understanding of how the interaction between thermal and acoustic conditions influences restorative outcomes in urban outdoor settings. These findings offer valuable insights for the evidence-based design and renewal of urban green spaces, particularly in contexts where environmental stressors such as heat and noise are increasingly prevalent.

Nevertheless, given the dynamic and complex nature of outdoor environments, two key limitations of this study must be acknowledged. First, although we assessed both the physical parameters of the thermal-acoustic environment and participants' comfort evaluations, we may have underestimated users' subjective soundscape preferences—such as perceived pleasantness and eventfulness—which could introduce potential bias in the interpretation of acoustic comfort. Despite this limitation, the findings remain valuable, as the *in-situ* approach captures the lived experiences of greenway users, offering a more holistic understanding of outdoor comfort and restorative potential. This, in turn, provides urban planners and policymakers with a richer evidence base for designing environments that support mental well-being. Second, the study focused exclusively on the hot summer period, without accounting for seasonal variations in thermal-acoustic comfort experiences. Future research should refine this limitation by incorporating seasonal dynamics—such as transitional periods and winter conditions—into an integrated framework that reflects the interplay between environmental and temporal characteristics of urban greenways.

Despite these limitations, the study makes a significant contribution to understanding how thermal-acoustic comfort in hot climates can enhance the restorative potential of urban greenways. It moves beyond traditional approaches that rely solely on physical environmental controls by introducing a perceptual perspective that accounts for the combined effects of thermal and acoustic experiences. These findings

highlight the potential of integrated sensory design strategies to improve restorative outcomes. Given the specific environmental characteristics of greenways, future studies should further explore the nonlinear relationships between thermal-acoustic conditions and psychological restoration. Doing so will support the development of more resilient and restorative outdoor spaces, particularly in response to sudden societal stressors and ongoing urban sustainability challenges [55].

6. Conclusions

Under the pressures of urban heat and rising climate stress, this study examined how thermal and acoustic conditions in outdoor spaces influence psychological restoration. The aim was to assess how combined thermal-acoustic comfort affects restorative outcomes in high-density urban contexts.

First, perceptual differences in thermal-acoustic comfort varied by greenway type: thermal conditions were more strongly linked to restoration in sparsely vegetated greenways, while acoustic conditions played a greater role in densely vegetated ones. Second, both TCV and ACV had distinct impacts under varying levels of thermal stress and noise. As thermal stress rose, TCV's influence on ROS increased in sparsely vegetated greenways. Conversely, in densely vegetated greenways, ACV remained positively associated with ROS even under higher sound pressure levels—highlighting the value of soundscape design in hot environments. Finally, a differentiated "veto effect" was observed: ACV showed stronger veto tendencies in densely vegetated settings, while TCV played a larger veto role in sparse ones. This suggests that enhancing acoustic comfort may be more effective in tree-rich environments, whereas improving thermal comfort is more critical in open, less-shaded spaces. The interaction between thermal and acoustic environments shapes psychological restoration differently depending on vegetation density. In the creation of comfortable urban outdoor spaces, improving thermal-acoustic conditions offers a promising pathway for enhancing restorative potential.

The comfort of urban greenways is closely linked to public health, particularly in high-density cities like Shenzhen, where greenway spaces directly impact residents' everyday experiences. Designing open spaces that consider both thermal and acoustic comfort can significantly enhance users' restorative outcomes, encouraging greater participation in outdoor leisure and mobility. By addressing the combined effects of thermal and acoustic environments, urban designers can more effectively create restorative urban landscapes through the targeted modulation of environmental comfort.

CRediT authorship contribution statement

Ming Gao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xun Zhu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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